

PTEC-SR

Superconducting Rotor - Proof-of-Concept White Paper

Concept status: exploratory engineering architecture, Version 0.1

Abstract

PTEC-SR is a proposed contactless rotating energy core built around superconducting magnetic bearing support, a magnetically coupled rotor, stator coils, vacuum or low-conductivity isolation, and a cryogenic subsystem. The purpose of the proof of concept is not to claim net energy creation, but to test whether very low mechanical bearing loss, stable high-speed rotation, and useful bidirectional electromagnetic coupling can coexist in a compact module. The rotor can store rotational energy, accept externally supplied mechanical or electromagnetic energy, and return part of that stored energy electrically through conventional induction.

1. Core Thesis

The PTEC-SR replaces conventional contact bearings with a superconducting magnetic bearing (SMB) arrangement. A levitated rotor is surrounded by stator coils used for motor/generator coupling. Thermal isolation separates the cold bearing region from any warm or plasma-facing region. In a larger VAPS/PTEC architecture, the rotor would function as an energy buffer and conversion stage rather than as a primary energy source.

Conservation boundary: Every electrical output term must be traceable to stored rotor energy or an external prime input. Magnetic levitation and superconductivity reduce losses; they do not supply continuous free energy.

2. Proposed Architecture

Subsystem	POC function
PTEC rotor core	Stores angular momentum; carries magnetic field structure.
Superconducting magnetic bearing	Provides contactless support and passive/active stabilization.
Stator coil array	Accelerates/brakes rotor and extracts electrical energy by changing magnetic flux.
Vacuum / low-conductivity enclosure	Reduces aerodynamic drag and thermal conduction.
Cryogenic loop	Maintains the superconducting bearing below its operating temperature.
Thermal barrier	Protects the cold stage from warm engine/plasma-side heat loads.
Power electronics	Controls motor/generator operation and interfaces with a DC bus/storage.
Sensors + controller	Monitors position, speed, temperature, vibration, field strength, and quench state.

3. Operating Principle

1) Cool the superconducting bearing assembly below its required operating temperature. 2) Establish the magnetic field configuration and levitate/center the rotor. 3) Use stator coils or another external prime mover to accelerate the rotor. 4) Hold rotational energy with minimized bearing and windage losses. 5) When power is requested, load the stator electrically; the resulting electromagnetic torque slows the rotor while generating electrical output. 6) Recharge by reversing the power flow and operating the stator as a motor.

The basic relationships are conventional: rotational energy $E = 1/2 I \omega^2$; mechanical power $P = \tau \omega$; and induced EMF follows the rate of change of magnetic flux. Useful round-trip efficiency is therefore bounded below 100% after stator, inverter, cryogenic, magnetic, aerodynamic, and control losses.

4. Initial POC Targets

Parameter	Exploratory target / test variable
Rotor geometry	Prismatic/tesseract-inspired geometry; mechanically balanced test article
Bearing	High-temperature-superconductor SMB or hybrid magnetic bearing
Environment	Vacuum or inert low-pressure enclosure
Air gap	Adjustable, approximately 2-5 mm starting design space
Rotation	Begin below 5,000 RPM; increase only after validated containment/balance
Electrical interface	Bidirectional inverter/rectifier connected to instrumented DC bus
Output class	Bench-scale first; characterize watts before setting kW targets
Primary metrics	Spin-down loss, bearing drag, cryogenic load, vibration, efficiency, quench margin

5. Test Program

Phase 1 - Static levitation: characterize lift, restoring force, thermal stability, and positional drift with a non-rotating mass.

Phase 2 - Low-speed rotor: validate balancing, emergency touchdown bearings, sensing, vacuum enclosure, and controlled spin-down.

Phase 3 - Loss mapping: measure decay rate versus pressure, temperature, RPM, and magnetic configuration to separate bearing, windage, eddy-current, and electrical losses.

Phase 4 - Motor/generator coupling: measure electrical input during acceleration and recovered electrical energy during controlled deceleration. Report round-trip efficiency from calibrated DC-bus measurements.

Phase 5 - Thermal integration: introduce representative external heat loads while maintaining cryogenic stability. A plasma-facing experiment, if pursued, should remain a separate later-stage subsystem until the rotor is independently validated.

6. Instrumentation and Falsifiable Success Criteria

Minimum instrumentation should include rotor RPM, three-axis displacement, vibration, stator voltage/current, DC-bus power, bearing and shield temperatures, vacuum pressure, cryocooler input power, and containment/interlock state. A useful POC succeeds if it demonstrates repeatable levitation, controlled acceleration/deceleration, quantified low-loss spin-down, stable thermal operation, and a measured energy balance that closes within instrument uncertainty.

A particularly important figure of merit is whole-system round-trip efficiency: recovered DC energy divided by DC energy used to charge the rotor, with cryogenic and control-system consumption explicitly included. This prevents a visually impressive low-friction rotor from being mistaken for an energetically favorable complete system.

7. Vehicle / VAPS Integration Hypothesis

Within the broader VAPS concept, PTEC-SR can be treated as a bidirectional buffer: external energy -> power electronics -> stator -> rotor, and rotor -> stator -> DC bus -> traction/storage. If a plasma or vapor subsystem is investigated, its contribution should be measured as an input or conversion process at the same system boundary. The PTEC-SR architecture does not by itself make a vehicle fossil-free; that depends on the upstream energy source and total lifecycle energy path.

8. Principal Engineering Risks

Risk	POC mitigation
Rotor burst / imbalance	Conservative RPM progression, precision balancing, remote operation, rated containment.
Superconductor quench	Temperature/field monitoring, automatic shutdown, energy dump path.
Magnetic instability	Active sensing/control and independent touchdown bearings.
Cryogenic parasitic load	Measure cryocooler input explicitly; optimize thermal isolation.
Eddy-current / hysteresis loss	Material selection and loss mapping across speed/field conditions.
Thermal ingress	Radiation shields, vacuum gap, ceramic/composite thermal breaks.
Overstated energy claims	Closed energy accounting with calibrated instruments and uncertainty bounds.

9. Decision Gate

The concept should advance only after the bench system demonstrates a favorable, reproducible loss budget and safe rotor dynamics. If cryogenic power plus electromagnetic losses overwhelm the bearing-loss advantage, the PTEC-SR may still be useful as a specialized flywheel or research platform, but not as an efficiency-improving vehicle energy system.

10. Conclusion

PTEC-SR is physically plausible as a superconducting-magnetic-bearing flywheel/generator research architecture. Its strongest testable proposition is not extraordinary energy production, but whether contactless support and controlled electromagnetic coupling can deliver sufficiently low losses to justify the added cryogenic, containment, and control complexity. The next rational artifact is a low-energy benchtop rotor with complete instrumentation and a closed energy ledger.

Document note: This white paper formalizes the PTEC-SR concept developed in the preceding POC discussion. Numerical performance values are intentionally treated as experimental targets rather than established results.